LECTURE 1

Introduction to DevSecOps

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LECTURE OVERVIEW

Learning Outcomes

1. Explain why DevSecOps is required.
2. Describe the limitations of traditional software development.
3. Explain the relationship between DevOps and security.
4. Describe the key principles and goals of DevSecOps.
5. Explain how security is integrated throughout the Software Development Life Cycle (SDLC).

What is DevOps?

WARM-UP

Imagine This Scenario...

A software company develops an online banking application.

Development	8 months
Testing	3 weeks
Security testing	Just before deployment

During the final security assessment, the team discovers:

- SQL Injection vulnerabilities
- Weak password storage
- Broken authentication
- Sensitive data exposure

The release is delayed by several weeks while developers fix these issues.

Discussion Question

Could these security issues have been prevented if security had been considered earlier?

MOTIVATION

Why Is Software Security Important?

Modern software stores and processes valuable information, including:

- Personal information
- Financial transactions
- Medical records
- Business data
- Government information

A SECURITY BREACH MAY RESULT IN

Financial losses

Loss of customer trust

Legal penalties

Service disruption

Damage to an organization's reputation

CONTEXT

Today's Software Landscape

Modern software systems are:

- **Cloud-based**
- **Web applications**
- **Mobile applications**
- **APIs**
- **Microservices**
- **Internet of Things (IoT)**
- **AI-powered systems**

As software becomes more connected, the number of potential attack surfaces increases.

Traditional Software Development

What is Software Development?

Software development is the process of designing, building, testing, deploying, and maintaining software systems to satisfy user requirements.

TYPICAL PHASES

01 Requirements Analysis

02 System Design

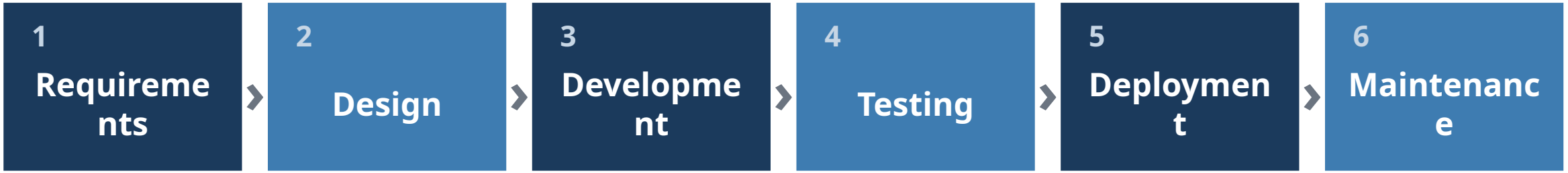
03 Development

04 Testing

05 Deployment

06 Maintenance

Traditional Software Development Life Cycle



Each phase completes before the next begins security is typically addressed only at the end, just before deployment.

Characteristics of Traditional Development

- Sequential process
- Teams work independently
- Manual testing
- Security checked near the end
- Long development cycles
- Slow customer feedback

Team Structure in Traditional Development



Each team works independently, often with limited communication.

DISCUSSION

Security Challenges in Traditional Development?

PART TWO

Introduction to DevOps

What is DevOps?

DevOps is a culture, set of practices, and collection of tools that enable collaboration between Development and Operations teams to deliver software faster and more reliably.



PART TWO

Objectives of DevOps

01

Faster software delivery

02

Continuous Integration (CI)

03

Continuous Delivery (CD)

04

Automation

05

Collaboration

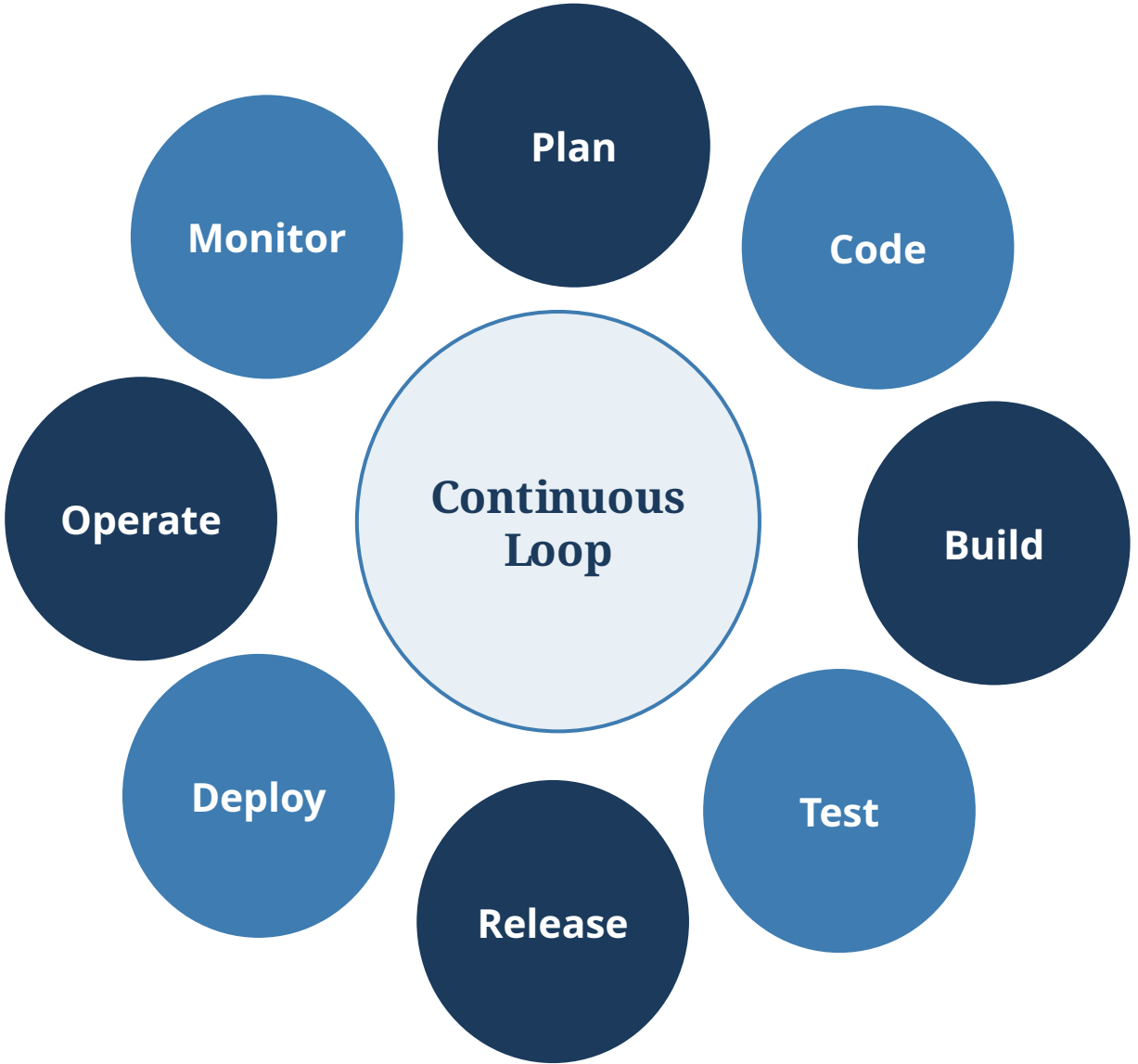
06

Continuous Monitoring

07

Higher software quality

DevOps Lifecycle



PART TWO

Benefits of DevOps

- Faster releases
- Reduced deployment failures
- Improved collaboration
- Automated processes
- Continuous feedback
- Improved software quality

TURNING POINT

Is DevOps Enough?

DevOps improves speed and collaboration.

However, security is often treated as a separate activity.

QUESTION

Can fast software still be insecure?

Yes.

This is exactly the gap DevSecOps is designed to close.

COMING UP NEXT

Introduction to DevSecOps

Bringing security into every stage of the software delivery lifecycle.

PART THREE

Introduction to DevSecOps

Integrating security into every stage of the Software Development Life Cycle.

PART THREE

What is DevSecOps?

DevSecOps is the practice of integrating security into every stage of the Software Development Life Cycle (SDLC). It combines Development, Security, and Operations into one collaborative process.



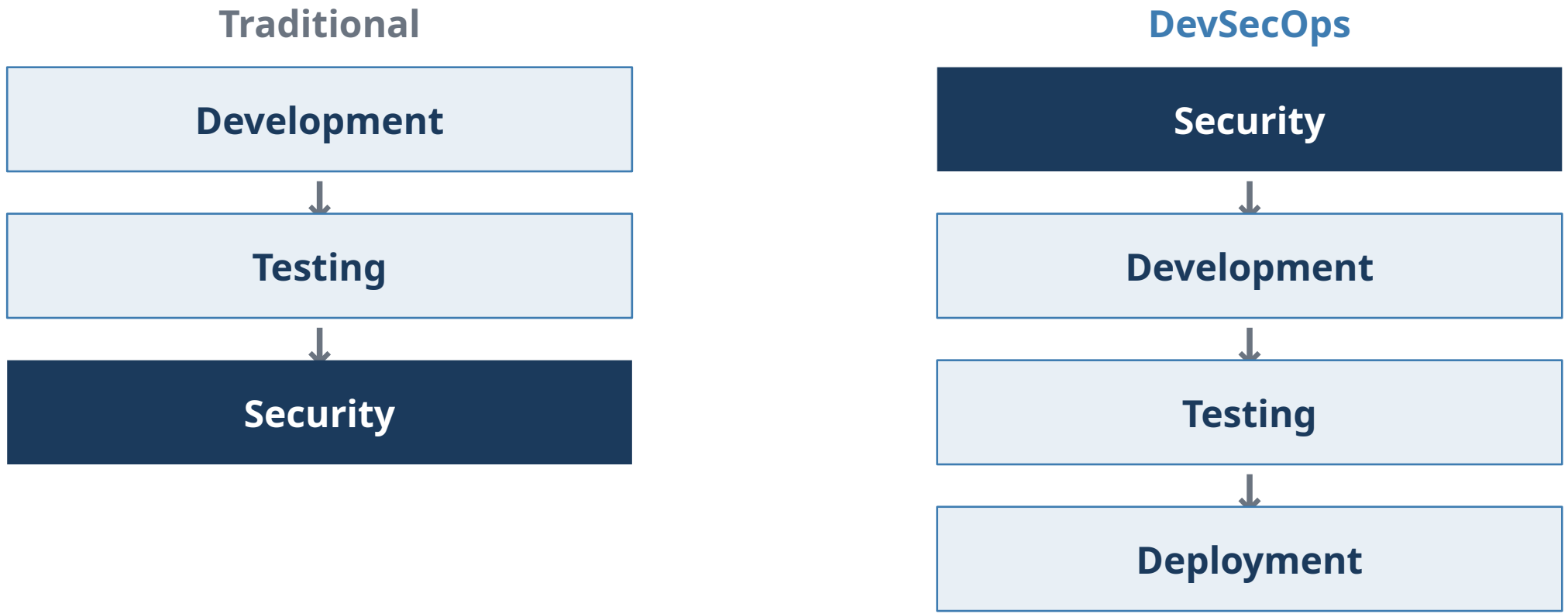
Security integrated across Development and Operations

PART THREE

DevOps vs DevSecOps

DevOps	DevSecOps
Focuses on speed	**Focuses on speed and security**
Security often added later	**Security integrated throughout**
Developers + Operations	**Developers + Security + Operations**
Limited security automation	**Continuous automated security**

Shift Left Security



Security begins from the earliest stages of development.

PRINCIPLES

Principle 1 – Security as a Shared Responsibility

Security is everyone's responsibility.

- **Developers**
- **Testers**
- **Operations Engineers**
- **Security Engineers**
- **Project Managers**

PRINCIPLES

Principle 2 – Automation

Automate repetitive security tasks such as:

- **Code scanning**
- **Dependency checking**
- **Vulnerability scanning**
- **Compliance verification**

Principle 3 – Continuous Security



Security should be continuously applied throughout the lifecycle.

PRINCIPLES

Principle 4 – Continuous Monitoring

Organizations should continuously monitor:

- **System logs**
- **User activity**
- **Security events**
- **Network traffic**
- **Threats**

PART THREE

Goals of DevSecOps

01

Deliver secure software

02

Detect vulnerabilities early

03

Reduce development costs

04

Improve collaboration

05

Increase automation

06

Achieve compliance

07

Build customer trust

APPLICATION

Integration of Security into the SDLC

SDLC Phase	Security Activity
Requirements	Security Requirements
Design	Threat Modeling
Development	Secure Coding
Build	Dependency Scanning
Testing	Security Testing
Deployment	Infrastructure Security
Operations	Continuous Monitoring

APPLICATION

Example: Online Shopping System

Requirements	Development	Testing	Deployment	Operations
■ User authentication ■ Secure payment gateway	■ Input validation ■ Password hashing	■ Vulnerability scanning ■ Penetration testing	■ Secure cloud configuration	■ Monitor login attempts ■ Patch vulnerabilities

SUMMARY

Key Takeaways

- ✓ Traditional software development introduces security too late.
- ✓ DevOps improves collaboration and software delivery.
- ✓ DevSecOps integrates security into every stage of development.
- ✓ Security becomes a shared responsibility.
- ✓ Automation enables faster and more secure software delivery.

DISCUSSION

Think–Pair–Share Activity

A hospital develops a patient management system.

Currently, security testing is performed only after the application has been completed.

Discuss in small groups:

- What risks could arise from this approach?
- How would DevSecOps improve the development process?
- Which SDLC phases should include security activities?

Be prepared to share your group's ideas with the class.

DISCUSSION QUESTIONS

What risks could arise from delaying security testing?

How would DevSecOps change this process?

Which phases need security built in?

Q & A

Questions and open discussion.